

PARALLEL PUMP STATION VER 1

Operator Training Simulator · cat-cracker feed pumps P-200A/B, T200, E-100, D-150

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CAUSE AND EFFECT DIAGRAM

DOCUMENT NO. : PPS-V1-IN-CE-001
DOCUMENT TITLE : CAUSE AND EFFECT DIAGRAM
P&ID : PPS-V1-PID-001 / PPS-PID-001
PFD : PPS-PFD-100
SRS : PPS-V1-IN-SRS-001
SHEETS : 1 (A3 landscape)

Teaching document generated from the simulator logic: bmsTrip(), motorTrip(), stopPump(), bmsStep(), tripConditions(), bmsCmd() reset rules, the start permissives and the alarm table ALM. Settings, effects and delays are checked against the model in the C&E matrix tab (section 5). Not for construction or operation of a real plant.

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
R0	09-Oct-26	Issued for teaching	Apinan Soottitantawat	—	—	

PARALLEL PUMP STATION VER 1 · P-200A/B CAUSE and EFFECT DIAGRAM Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University								CLIENT :		Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University							REV	DATE	ORIGINATOR	CHECKER	APPROVER		
								PROJECT :		Parallel Pump Station Ver 1 · OTS (P-200A/B)							R0	09-Oct-26	Apinan Soottitantawat	—	—		
								DOCUMENT No. :		PPS-V1-IN-CE-001													
								DOCUMENT TITLE :		Cause and Effect Diagram													
								P&ID / PFD :		PPS-V1-PID-001 / PPS-PID-001 / PPS-PFD-100												SHEET 1 of 1	
CAUSES								EFFECTS												Notes: 1. Action codes: STOP = motor stopped by the MCC contactor; C = close (V100: FRC-100 output forced to 0 %, the controller tracks); TP = tripped and latched (manual reset on the SIS tab); F = first-out; A = activate; P = permissive (the pump cannot be started while the condition holds); RB = reset blocked; R = acts only with the Rev C MOV interlock in service. 2. Rev C option: when the MOV interlock is in service every stop of a pump also drives its discharge MOV (V201 or V205) shut; a start reopens it 5 s after the motor starts. Without Rev C the hand valves stay as they are. 3. LSL-200 is armed only while a pump runs. It reads LI-200, the same measurement as the LI-200 alarm and the LSL-200 start permissive (Safety layers tab, independence check). 4. ESD: no delay. It stops both pumps, latches the unit trip with the ESD as first-out, forces FRC-100 to 0 % (V100 shut) and drives TV-110 shut until reset; pump starts are refused while it is active. It is the only cause that cuts the steam. 5. Delays are counted in the logic solver at the 0.25 s scan; a trip acts at the first scan after its delay has passed (5.25 s and 10.25 s in the model). The 49 timers count only while the motor runs above 80 % speed, so the starting current of a motor does not trip it. 6. TSHH and 49 trips latch per motor (no unit trip, no first-out): only that pump stops; the other pump is not affected. 7. LSHH-150 latches and forces V100 shut but does not stop the pumps: the running pump is then deadheaded until TSHH-212/222 stops it (HAZOP dynamic check D7). 8. LALL-150 belongs to the cat-cracker unit (neighbour): it stops the cat-cracker charge pump and the station shows FI-151 LOST. It is cleared when the cat cracker restarts its charge. 9. Alarms have no delay; they clear with a 1 % deadband. Every new alarm sounds the horn until silenced and enters the alarm list for acknowledgement. 10. Suppression by plant state: FRC-100 alarms only while the cat cracker draws its charge; the TI-110 LO alarm only while feed flows (above 3 L/s). 11. PAH-210/220 exist only with the Rev C casing pressure transmitters PIT-210/220; the alarm is armed while that pump is stopped. 12. LSL-200 is a sequence permissive, not a trip: no valve or motor moves; a start command is refused while T200 is below the setting. 13. Reset of the SIS and motor latches (one command on the SIS tab) is refused while T200 is below the LSL-200 setting after a unit trip, a pump casing is above the setting, a protection relay is still tripped, motor A is still single-phasing, or D-150 is above the setting after LSHH-150. One reset clears the unit trip, both motor latches, the LSHH-150 latch and the ESD. 14. Instrument-air failure is not an SIS action: V100 and TV-110 go to their fail-closed positions. Power failure stops both motors (de-energise to trip). The seal-leak alarm of each pump (gas detector) is annunciated too; it has no setting in the simulator and is not listed as a cause.			
								STOP	STOP	C	C	C	C	TP	F	TP	TP	TP	STOP				
Item No.	Tag No.	Tag Service Description	Trip/Alarm Direction	Setting	Eng. Unit	Delay	Note No.																
001	HS-ESD	Emergency shutdown push-button (SIS tab)	Pushed	—	—	0 s	1, 4	X	X	R	R	X	X	TP	F					X		X	
002	LSLL-200	T200 level low-low on LI-200 while a pump runs: pumps losing suction (unit trip)	Low-Low	10	%	5 s	2, 3, 5	X	X	R	R			TP	F					X		X	
003	TSHH-212	Pump A casing temperature high-high on TI-212 (deadhead, dry running)	High-High	110	°C	5 s	2, 6	X		R						TP				X		X	
004	TSHH-222	Pump B casing temperature high-high on TI-222	High-High	110	°C	5 s	2, 6		X		R					TP				X		X	
005	49-210	Motor A overload relay on II-210 (running above 80 % speed)	High	21.7	A	10 s	2, 5, 6	X		R						TP				X		X	
006	49-220	Motor B overload relay on II-220 (running above 80 % speed)	High	21.7	A	10 s	2, 5, 6		X		R					TP				X		X	
007	LSHH-150	D-150 level high-high on LI-150: V100 forced shut (pumps keep running)	High-High	95	%	0 s	7					X					TP			X		X	
008	LALL-150	D-150 level low-low: cat-cracker charge pump tripped (cat-cracker logic)	Low-Low	10	%	0 s	8											X	X			X	
009	FRC-100	Feed flow to the cat cracker low (FT-100)	Low	12.6	L/s		9, 10														X	X	
010	FRC-100	Feed flow to the cat cracker high (FT-100)	High	17	L/s		9, 10														X	X	
011	TI-110	E-100 outlet temperature low	Low	140	°C		9, 10														X	X	
012	TI-110	E-100 outlet temperature high	High	165	°C		9, 10														X	X	
013	LI-200	T200 feed storage level low	Low	20	%		9														X	X	
014	LI-200	T200 feed storage level high	High	85	%		9														X	X	
015	LI-150	D-150 cat-cracker feed drum level low	Low	30	%		9														X	X	
016	LI-150	D-150 cat-cracker feed drum level high	High	80	%		9														X	X	
017	II-210	Motor A current high (MCC)	High	20	A		9														X	X	
018	II-220	Motor B current high (MCC)	High	20	A		9														X	X	
019	TI-212	Pump A casing temperature high (pre-alarm of TSHH-212)	High	85	°C		9														X	X	
020	TI-222	Pump B casing temperature high (pre-alarm of TSHH-222)	High	85	°C		9														X	X	
021	PAH-210	Pump A casing pressure high while A is idle (PIT-210, Rev C option)	High	8	m		9, 11														X	X	
022	PAH-220	Pump B casing pressure high while B is idle (PIT-220, Rev C option)	High	8	m		9, 11														X	X	
023	LSL-200	Pump start permissive on LI-200 (suction covered); also blocks the reset of a unit trip	Low	15	%		12, 13	P	P					RB									
024	TI-212/222	Reset permissive: a pump casing still hot blocks every reset	High	85	°C		13							RB		RB	RB	RB					
025	LI-150	Reset permissive after LSHH-150: D-150 still high blocks every reset	High	90	%		13							RB		RB	RB	RB					
026																							
027																							